DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #03E: MARTIAL ARTS & TECHNIQUES (PART 5 — DEFENSIVE KUNGFU, PROTECTIVE AEGIS, PASSIVE SLOTS, INNATE SUBPATH ARTS & PASSIVE CONSTELLATIONS)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Series Context**:
  - `03A`: Combat Moves (`COMBAT_TECHNIQUE` / `moveDefs`), `SK.list` Cache, `chooseSkill` AI, 16 Weapon Mismatches & Spatial Range.
  - `03B`: Composable Keyword Engine, 34 Elemental Reactions (`DARK_YIN` vs `YIN_SHADOW`), & Duet Synergies.
  - `03C`: Mantras & Inner Qi Circulation (`MANTRA` / `MANTRA_SYSTEM`), Lifespan Traps & Sect Formation Warfare.
  - `03D`: Qinggong, Sword Flight, Celestial Levitation, Stealth/Recon Arts & Combat Mobility.
  - **`03E` (This Document)**: Dedicated, exhaustive deep dive into **Defensive Kungfu & Protective Aegis (`DEFENSIVE_TECHNIQUE` / `CULT.DEFENSIVE_ARTS` — 64 manuals), Passive Arts (`PASSIVE` — 227+ manuals & `passiveLoadout`), Innate Subpath Arts (`INNATE_SUBPATH` / `M.activeInnateTechs` across 286 subpaths), and Passive Resonance Constellations (`CULT.PASSIVE_RESONANCE` — 6 Constellations)**.

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **40-Script Load Order**: Preserve exact `<script>` load sequence in `index.html` and `test/load.js`.
3. **100% Emergent NPC Autonomy**: Every Defensive Guard Posture (`PASSIVE_AURA_GUARD`, `COUNTER_PARRY_STANCE`, `FORTRESS_BASTION_LOCK`), Passive Loadout selection, Life-vs-Combat Passive prioritization, and Passive Constellation Focus (`setFocusConstellation`) must be initialized and adapted autonomously by NPCs—never waiting for manual UI button clicks.

4. **Test-Lock Awareness**: Note that `test/v560_defensive_kungfu_and_aegis_suite.js:68` asserts `allDefTechs.length === 58` (for `JD.TECH.techniques` with `cat === 'DEFENSIVE_TECHNIQUE'`) and `test/v580_forbidden_arts_and_passive_constellations_suite.js:76-78` asserts `227 PASSIVE + 36 INNATE_SUBPATH = 263`. Enrich runtime lookup maps (`defensiveArchetype`, `passiveConstellation`) via non-destructive metadata properties rather than mutating locked `t.cat` counts!

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1. [DEF-CAT-01] Defensive Kungfu Catalog Schism: Iconic Shield/Bell/Armor Arts Excluded From `CULT.DEFENSIVE_ARTS`

1.1 Over 25 Pure Defensive Manuals in `data_tech.js` Have Zero Access to `CULT.DEFENSIVE_ARTS`

- **Locations**: `data/data_tech.js`, `js/cultivation.js` (`CULT.DEFENSIVE_ARTS`), `test/v560_defensive_kungfu_and_aegis_suite.js`
- **Root Cause**:
  - Patch `v5.6.0` only attached `defensiveArchetype` / `defensiveSpec` to the 58 newer manuals whose `cat === 'DEFENSIVE_TECHNIQUE'` plus 6 `DEF_*` signature manuals.
  - Meanwhile, in `data_tech.js`, **all 120 core `ART190_*_001..040` arts** were authored with `cat: "COMBAT_TECHNIQUE"` and `role: "COMBAT_DEFENSE"`—including **pure defensive shield, armor, parry, and bodyguard scriptures**:
    - `ART190_XIAN_010` (เคล็ดปราณเกราะหยกพันชั้น — Thousand-Layer Jade Armor)
    - `ART190_XIAN_026` (เพลงกระบี่สะท้อนเงาวารี — Water-Mirror Reflection Sword)
    - `ART190_XIAN_036` (เคล็ดปราณสายฟ้าคุ้มตันเถียน — Lightning Dantian Ward)
    - `ART190_BODY_003` (ฝ่ามือระฆังทองวัชระสะท้อนศึก — Golden Bell Vajra Reflection)
    - `ART190_BODY_012` (เคล็ดกายาสำริดต้านคมกระบี่ — Bronze Skin Blade-Ward)
    - `ART190_BODY_022` (วิชาโล่เนื้อปราการเหล็กคุ้มสหาย — Flesh-Shield Iron Bastion for Allies)
    - `ART190_BODY_036` (วิชากายาหนามเหล็กสะท้อนแผล — Iron-Thorn Reflection Body)

    - `ART190_BODY_038` (เคล็ดกายาหินผาไม่หวั่นคลื่นลม — Unshakeable Cliff Body)
    - `ART190_GOD_015` (เคล็ดเกราะแสงศรัทธาป่วงชน — Faith Light Shield)
    - `ART190_GOD_027` (เคล็ดเทวรูปศิลาไร้ความเจ็บปวด — Painless Stone Idol Ward)
    - `ART_V112_DEMON_BONE_ARMOR_SHIELD_T3` (เคล็ดเกราะกระดูกขาวอเวจี — `combatArch: "guard"`)
    - `ART_SOUL_MIRROR_COUNTER_T2` & `ART_V112_SOUL_VOID_LOTUS_REFLECT_T4` (ดอกบัวสุญตาสะท้อนหมื่นธรรม)
  - **Result**: A `BODY` tank who learns วิชาโล่เนื้อปราการเหล็กคุ้มสหาย (`ART190_BODY_022`) or วิชากายาหนามเหล็กสะท้อนแผล (`ART190_BODY_036`) **cannot activate `FORTRESS_BASTION_LOCK` or `ELEMENTAL_THORN_ARMOR`**, because `CULT.DEFENSIVE_ARTS` does not recognize them as defensive techniques!
- **Fix**: Keep `t.cat` unchanged (preserving `v560`'s `58` count), and attach `t.defensiveArchetype` + `t.defensiveSpec` to any `role === 'COMBAT_DEFENSE'` technique whose `combatArch` is `'guard' | 'shield' | 'counter'` or whose Thai title matches `/เกราะ|โล่|ระฆังทอง|คุ้มกาย|คุ้มตันเถียน|คุ้มสหาย|ปราการ|สะท้อน|ต้านคม|หนามเหล็ก/`.

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2. `[DEF-CMB-02]` Five Live Runtime Bugs in `CULT.DEFENSIVE_ARTS` (`combat.js:310–355` & Spec Drift)

2.1 Bug 2A — 100% Dormant in Autonomous Play (`setDefensivePosture` Only Called by `ui.js`!)

- **Locations**: `js/combat.js:312, 327` (`if (W._hasActiveDefSupportAura)` and `if (d.defensiveArtState && d.cult)`), `js/cultivation.js`
- **Verified Fact**: Searching the entire repository for `setDefensivePosture` returns **only `cultivation.js`, `ui.js`, and `v560` test**.
- **Consequence**: When an NPC studies `DEF_TAIJI_STAR_SHIFT_T3` or `DEF_IMMORTAL_HEAVEN_AEGIS_T6` from the Scripture Pavilion, **`d.defensiveArtState` remains `undefined` and `W._hasActiveDefSupportAura` remains `false`**! Unless the player manually opens `UI.rNpc` for every single NPC and clicks a posture button, **all 64 Defensive Kungfu manuals provide `0%` `DEFENSIVE_ARTS` mitigation, `0%` Taiji redirection, `0%` Cicada Molt, and `0%` Ally Aegis Interception in real gameplay**!

2.2 Bug 2B — Infinite Qi Battery & 25-Tile Laser Counter-Slashes on Poison/Bleed DoT Ticks (`combat.js:327–348`)!

- **Exact Code (`combat.js:327–347`)**:

```javascript
const rawBefore = dmg;
dmg = Math.max(1, Math.round(dmg * (1 - df.effDmgRedPct)));
...
if (d.h && d.h.state && df.qiPerHit > 0) d.h.state.qi = Math.max(0, (d.h.state.qi
|| 0) - df.qiPerHit);
if (df.effRedirectPct > 0 && a && C.alive(a)) {
  const redirDmg = Math.max(1, Math.round(rawBefore * df.effRedirectPct));
  if (d.h && d.h.state && df.spec.qiRestoreOnHit) {
    d.h.state.qi = Math.min(CULT.qiMax(d), (d.h.state.qi || 0) +
df.spec.qiRestoreOnHit);
  }
  C.applyDamage(W, f, d, a, redirDmg, 'สะท้อนวิชาคุ้มกาย');
} else if (df.effCounterPct > 0 && a && C.alive(a)) {
  const cntDmg = Math.max(1, Math.round(rawBefore * df.effCounterPct));
  C.applyDamage(W, f, d, a, cntDmg, 'สะท้อนวิชาคุ้มกาย');
}
```

- **Four Simultaneous Bugs in That Block**:
  1. **Poison/Bleed DoT Exploit (`Infinite Qi Battery + Cross-Map Counter`)**: Every minute `C.statusTick` (`combat.js:959`) ticks `poison`/`bleed`/`burn` on `d`, passing poisoner `src` as `a`. On `TAIJI_SOFT_REDIRECTION`, **every poison tick inside `d`'s veins triggers "☯️ ปราณอ่อนยืมแรงสะท้อนกลับ!"**, recovering `+5 Qi` (`qiRestoreOnHit > qiPerHit` $\to$ **being poisoned becomes a net positive Qi generator!**) and shooting deflected force across the entire map onto `src`!
  2. **Missing Melee Distance Check on `SWORD_CURTAIN_WARD` & `ELEMENTAL_THORN_ARMOR`**: `ACTIVE_PATCH_PROMPT_V5_6_0.md` explicitly specified *"automatically counter-slashes **melee** attackers for 16% damage"*, yet line 343 has **zero distance check**, so a spinning 1.5-meter Sword Curtain slashes archers and spellcasters standing 25 tiles away!
  3. **Pre-Mitigation `rawBefore` & Uncapped Reflection**: Using `rawBefore` (before `effDmgRedPct`) and calling `C.applyDamage` directly bypasses `COMBAT.reflectOf`'s unified `35% dmg / 22% attacker MaxHP` ceiling!
  4. **Multi-Hit Qi Wipeout**: A 5-hit low-multiplier flurry deducts `5 × df.qiPerHit` in one action without per-cast ICD scaling.

2.3 Bug 2C — `SHADOW_SAMURAI` (`qiMax === 0`) Locked Out by `df.qiSufficient`!

- **Location**: `js/combat.js:329` (`if (df && df.qiSufficient)`) & `js/cultivation.js` (`resolveNpcDefense`)
- **Defect**: `SHADOW_SAMURAI` has dedicated defensive techniques (`TECH_V230_SHADOW_SAMURAI_DEFENSE` — วิชาคุ้มกายซามูไรไร้เงา), yet `qiMax(d) === 0` and `qi === 0`! Checking `qi >= qiPerHit` or `qi >= qiMax * 0.20` (`CELESTIAL_AEGIS_DOME`) fails on Zero-Qi Samurai unless `CULT.isZeroQiSamurai(d)` substitutes Stamina (`h.state.energy >= 12`, deducting `0.35 energy` per parried strike)!

2.4 Bug 2D — Stacking Up to 7 Cheat-Death Saves in One Fight (`Cicada Molt` + `Substitution` + `Nova` + `Phoenix` + `Decoy` + `Karma` + `Last Breath`)!

- **Location**: `js/combat.js:349–480` (`C.applyDamage`)
- **Emergent Balance & Context Issue**:
  - Inside `C.applyDamage`, **nine separate survival/cheat-death interceptors** sit in sequence without a shared per-fight survival-save cap:
    1. `MIRAGE_CICADA_SHELL` (`<= 25% HP`, negates blow + heals `22%`)
    2. `SHADOW_CLONE` (`negates 3 full hits`)
    3. `SUBSTITUTION` (`<= 0 HP`, negates blow + heals `25%`)
    4. `TABOO_DESPERATION_NOVA` (`<= 30% HP`, absorbs `65%` + heals `25%`)
    5. `VERMILION_PHOENIX` Pet (`<= 15% HP`, restores to `42%` + `25%` shield)
    6. `MECH_DECOY` / `KARMA_ECHO` / `NIRVANA_GRACE` (`<= 12% HP`, restores to `35–38%` + shield)
    7. `PHOENIX_CLAUSE` (`<= 0 HP`, restores to `48%` + `22%` shield)
    8. `KW_LAST_BREATH` (`<= 0 HP`, survives at `1 HP` + `22%` shield)
  - An late-game Elder combining a Defensive Cicada manual, Forbidden Art, Phoenix Pet, Nirvana buff, and `KW_LAST_BREATH` requires **6+ consecutive "lethal" blows** after reaching 0% HP just to go down once!
  - **Fix**: Introduce a clean `fTac.majorCheatDeathsUsed[d.id]` counter capped at **at most `2` full death-negation/rebirth saves per fight** (excluding ordinary shields), applying `50%` diminishing heal efficiency on the 2nd save!

2.5 Bug 2E — Unwired Spec Promises on `IRON_VAJRA_CARAPACE` & `SWORD_CURTAIN_WARD`

- **Compare `ACTIVE_PATCH_PROMPT_V5_6_0.md` vs Live Code**:
  1. `IRON_VAJRA_CARAPACE` promised **`+45%` Poise shield against stagger/channeling interruption** (`startChargedSkill` in `combat.js:858,`

`1689` hardcodes `poiseShield` without multiplying by `1.45` when `IRON_VAJRA_CARAPACE` is active, and non-charged `channel` interruption at `dmg >= hm * 0.12` in `combat.js:461` ignores `IRON_VAJRA_CARAPACE`!).

2. `IRON_VAJRA_CARAPACE` promised **`-65%` daily work/forging/gathering injury risk** (`AI.injure` in `ai.js` does not check `IRON_VAJRA_CARAPACE` when rolling work/spar/forge accidents!).
3. `SWORD_CURTAIN_WARD` promised **`+25%` Sect Guard/Patrol security** (`A.guard` in `ai.js` never checks `resolveNpcDefense`!).

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3. `[PASS-SLOT-03]` Passive Loadout Structural Flaws: Flat 3-Slot Cap Across 9 Realms, Life-vs-Combat Slot Cannibalism & CP Waste

3.1 Life-Skill Passives (`passive.life`) Cannibalize the 3 Combat `passiveLoadout` Slots!

- **Locations**: `js/martial.js` (`M.reconcilePassives`, `M.passiveStats`, `MAX_PASSIVE_SLOTS = 3`), `js/cognition.js` (`COG.reconcileLoadout`)
- **Design & Contextual Flaw**:
  - Over 60 profession/subpath passives (`PASS_V214_*`, `PASS_V234_*`, craft/herb/medicine/scripture passives) grant life-skill bonuses via `t.passive.life` (`ALCHEMY`, `FORGING`, `HERBALISM`, `MEDICINE`, `ARRAY`, `TALISMAN`, `SCRIPTURE`).
  - However, `M.passiveStats(n)` **only reads `passive.life` from the 3 active slots in `n.passiveLoadout`!**
  - **Chain-Reaction Consequence**:
    1. When an Alchemist or Smith Elder learns higher-tier combat passives (`tier 5 +ATK / +DEF`), `M.reconcilePassives` (which sorts purely by `tier` and `power`!) **kicks their Tier-3 `+ALCHEMY` / `+FORGING` passive out of `n.passiveLoadout`**, instantly stripping the Grandmaster Alchemist of their alchemy mastery!
    2. Conversely, if they keep 2 craft passives pinned in `passiveLoadout`, they enter lethal Sect Raids with only **1** combat passive slot!
- **Second Structural Flaw — Flat `3` Passive Slots From Stage 1 Mortal Initiate to Stage 9 Immortal Sovereign**:
  - A Stage-1 Qi-Condensation novice and a 1,500-year-old Stage-9 Immortal Sovereign both have **`3` passive slots**, even though `CULT.PASSIVE_RESONANCE` (`v5.8.0`) defines **Constellation Set Tier 3 at `4+` matching passives**!

- **Third Flaw — `COG` Keeps Buying Redundant Passives After All Slots Are Full (`cognition.js:330`)**:
    - `COG.techPreference` gives a flat **`+16` bonus to every `PASSIVE` manual** regardless of whether `n.passiveLoadout` is already full of Tier 5–6 passives! Disciples waste hundreds of CP studying Tier-2/Tier-3 passives that immediately sit dead and unslotted forever!

3.2 Remediation Blueprint (`[PASS-SLOT-03]`)

1. **Separate Craft/Auxiliary Mind-Slots (`n.lifePassiveLoadout`, max `2` slots) From Combat Passive Slots (`n.passiveLoadout`)**:
    - Passives whose primary payload is `passive.life` (and `0` combat stats) go into `n.lifePassiveLoadout` (`2` slots), never competing with combat survival passives! Hybrid passives (`combat + life`) prioritize matching the NPC's appointed `n.post` (`alchemist`, `smith`, `healer`, `scholar`, `arraymaster`).
2. **Realm-Scaled Combat Passive Capacity (`2 -> 3 -> 4 -> 5` Slots)**:
    - Scale `M.maxPassiveSlots(n)` smoothly by cultivation realm: Stage 1–2: `3` slots (preserving early-game test parity!), Stage 5–6: `4` slots (allowing true **4-piece Active Constellation Tier 3**!), Stage 8–9: `5` slots!
3. **Marginal-Gain Scoring in `COG.techPreference`**:
    - Score a candidate `PASSIVE` manual by **`marginalDelta = score(t) - worstSlottedPassiveScore(n)`**! If `t` cannot beat any currently slotted passive (and does not advance the NPC's focused Constellation threshold), drop its `+16` preference bonus to `+1`, redirecting the NPC's CP toward pills, breakthroughs, or upgrades!

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4. `[PASS-PIPE-04]` Stat & Condition Pipeline Bugs in Passives (`n._hpf` Denominator & `cnd.p.matk`)

4.1 `n._hpf` Uses Naked `hpMaxBase` Instead of Actual `CULT.hpMax(n)` (`js/stats.js`)

- **Locations**: `js/stats.js` (`compute` & `getBaseFast` calculation of `n._hpf`), `js/martial.js` (`M.passiveStats` evaluating `hpBelowPct` / `hpAbovePct`)
- **Mathematical Bug**:
    - `stats.js` computes `n._hpf = n.h.state.hp / hpMaxBase`, where `hpMaxBase` is the **raw unbuffed stage HP** *before* `+hp%` from armor, constitutions, meridians, mantras, and passives!
    - **Example 1 (Tank With `+110% HP` Gear/Body)**: `hpMaxBase = 1,000`, actual `CULT.hpMax(n) = 2,100`. When wounded to **`1,000 / 2,100`** (**`47.6%` real**

HP), `n._hpf` evaluates `1,000 / 1,000 = 1.00` (**100%**)! Every `hpBelowPct: 0.50` bastion/berserk passive **fails to activate** until the tank drops below `500 / 2,100` (`23.8%` real HP)!
    - **Example 2 (Glass-Cannon / Blood-Pact With `-15% HP`)**: Even at **100% full health (`850 / 850`)**, `n._hpf = 850 / 1,000 = 0.85`, permanently disabling `hpAbovePct: 0.90` full-health passives!
- **Fix**: Compute `n._hpf` from the final resolved `hp` cap (`clamp(n.h.state.hp / Math.max(1, prevFinalHpMax || hpMaxEst), 0, 1)`) using a two-pass or cached `n._lastHpMax` denominator, and call `ST.dirty(d)` whenever `C.setHp` crosses a `0.90`, `0.70`, `0.50`, or `0.35` threshold!

4.2 Target-Conditional Spellcaster Passives (`cnd.p.matk`) Grant `+0%` Damage (`js/martial.js:399`)

- Ensure `M.combatTargetBonus(a, d, attackType)` (referenced in `03B` Section 4.2) applies `cnd.p.matk` to magical (`'m'`) moves and `cnd.p.atk` to physical (`'p'`) moves, plus flat `cnd.f` (`crit`, `critDmg`, `pen`, `acc`).

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5. `[INNATE-CONST-05]` 87% of Subpaths Excluded From `CULT.PASSIVE_RESONANCE` & Zero Autonomous Constellation Focus

5.1 Only 36 of 286 Subpaths Have `cat === 'INNATE_SUBPATH'` (`v580` Blindspot!)

- **Locations**: `js/cultivation.js` (`CULT.PASSIVE_RESONANCE`), `data/data_tech.js`, `data/data_martial.js`, `test/v580_forbidden_arts_and_passive_constellations_suite.js:76-81`
- **Root Cause**:
    - Across 4 development waves, Subpath Innate Arts (`M.activeInnateTechs(n)`) were authored under different `cat` strings:
        - Wave 1 (`ART_C117_INNATE_*`) & Wave 2 (`INNATE_*_CORE`) & Wave 4 (`INNATE_V200_*` — **154 Grand Dao subpaths!**) used `cat: 'COMBAT_TECHNIQUE'` (or `'SOUL_MANTRA'` / `'FORBIDDEN_ART'`).
        - **Only Wave 3 (`INNATE_V234_*` — 36 subpaths)** used `cat: 'INNATE_SUBPATH'`!
    - Because `v5.8.0` filtered `t.cat === 'PASSIVE' || t.cat === 'INNATE_SUBPATH'`, **250 out of 286 subpaths (`87.4%` of all subpaths in the**

game!) do NOT have `passiveConstellation` on their Subpath Innate Art, putting them a full star behind the 36 `v2.0.34` subpaths in unlocking Constellation Set Tiers!

- **Second Flaw — Counting Unslotted Library Bloat Instead of Active Build**:
  - `PASSIVE_RESONANCE` counts **every `PASSIVE` ever pushed into `n.cult.techs`**, even Tier-1 passives that aren't in `n.passiveLoadout` and fail `M.conditionMatches`!
- **Third Flaw — Zero Autonomous Calls to `setFocusConstellation`**:
  - As verified by repository search, `setFocusConstellation` is called **only in `ui.js` and `v580` test**. Without player clicks, `n.passiveResonanceState` is `undefined` on 100% of NPCs!

5.2 Remediation Blueprint (`[INNATE-CONST-05]` & Autonomous Defensive Postures)

1. **Enrich All 286 Subpath Innate Arts With `passiveConstellation`** (without altering `t.cat`, so `v580`'s `36 INNATE_SUBPATH` assertion continues to pass 100%!) and count **active `passiveLoadout` + `lifePassiveLoadout` + `M.activeInnateTechs(n)`** (plus `0.5` resonance credit for unslotted mastered passives `tm >= 40`) toward Constellation Tiers!
2. **Autonomous Constellation Alignment (`CULT.PASSIVE_RESONANCE.autoSyncConstellation(W, n)`)**:
   - Call inside `M.reconcilePassives` / `COG.reconcileLoadout`: automatically select the Constellation with the highest active count (breaking ties by Path/Subpath stat affinity), populating `n.passiveResonanceState` organically for every cultivator!
3. **Autonomous Defensive Guard Posture AI (`CULT.DEFENSIVE_ARTS.evaluateAutonomousPosture(W, n, f)`)**:
   - Auto-initialize `n.defensiveArtState` as soon as `n` learns any defensive technique (setting `W._hasActiveDefSupportAura = true` when applicable).
   - **Out of Combat**: Default to `'PASSIVE_AURA_GUARD'` (`0.5×` Qi cost, `-65%` work injury protection).
   - **In Combat (`COMBAT.enter` / `fightStep`)**:
     - Switch to **`'FORTRESS_BASTION_LOCK'`** if `C.inferCombatRole(n) === 'TANK'` or `n` is protecting a wounded `companionId` / `mentor` / `CASTER` ally within `5.5` tiles (`loyalty >= 60` or `empathy >= 60`).
     - Switch to **`'COUNTER_PARRY_STANCE'`** for Duelists, Sword Immortals, Taiji practitioners, or solo fighters (`1v1` duels).
     - Fall back to **`'PASSIVE_AURA_GUARD'`** when `qi < qiMax * 0.25` to preserve Qi for active skills!

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6. Implementation & Verification Checklist for GPT / Gemini (03E)

- [ ] **Defensive Catalog & Spec Parity**: Attach `defensiveArchetype` / `defensiveSpec` to legacy `ART190_*` shield/armor/guard arts (preserving `cat` counts); wire `IRON_VAJRA_CARAPACE` (`+45%` poiseShield, `-65%` work/forge injury risk, `-6%` eva) and `SWORD_CURTAIN_WARD` (`+25%` `A.guard` security).
- [ ] **Combat Defensive Safety (`combat.js:310–355`)**: Block `DEFENSIVE_ARTS` reflection/Qi-restore from triggering on `statusTick` DoTs and `tickDomain`; enforce melee distance (`<= 2.5` tiles) on `SWORD_CURTAIN_WARD` & `ELEMENTAL_THORN_ARMOR` counters; compute redirection from post-mitigation `dmg` inside the unified `<= 22% MaxHP` reflection cap; support Stamina upkeep for Zero-Qi `SHADOW_SAMURAI`; cap full cheat-death saves at `2` per unit per fight.
- [ ] **Passive Slots & Condition Math**: Split `lifePassiveLoadout` (`2` craft slots) from `passiveLoadout` (`3..5` realm-scaled combat slots); add marginal-gain passive scoring in `COG.techPreference`; fix `n._hpf` in `stats.js` to divide by actual resolved `hpMax` instead of raw `hpMaxBase`.
- [ ] **Innate & Constellation Autonomy**: Enrich all 286 subpath innate arts (`^INNATE_|^ART_C117_INNATE_`) with `passiveConstellation` metadata; auto-run `autoSyncConstellation` and `evaluateAutonomousPosture` so 100% of NPCs activate their Defensive Aegis and Passive Constellations naturally without UI clicks!